QUALIFICATION REQUIREMENTS REMOVED

MIL-M-38510/18A
9 August 1983
SUPERSEDING
MIL-M-38510/18(USAF)
4 September 1975

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, REGISTER FILE, MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, register file microcircuits. One product assurance class and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, with the exception that the "JAN" or "J" certification shall not be used.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	4 x 4 register files

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -12 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation per flip-flop, (P_D) 1/	770 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Case E	0.08°C/mW
Case F	0.06°C/mW
Junction temperature (T_J)	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5962

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1.4 Recommended operating conditions.

Supply voltage	+4.5 V dc minimum to +5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	+2.0 V dc
Maximum low-level input voltage (V_{IL})	+0.8 V dc
Normalized fanout (each output) 2/	10 maximum
Case operating temperature range	-55°C to +125°C
Input setup time 3/:	
Data input	10 ns minimum
Write select 4/	15 ns minimum
Input hold time 3/:	
Data input	15 ns minimum
Write select 4/	5 ns minimum
Latch time for new data 5/	25 ns minimum

1.5 Description. The 16-bit TTL register file incorporates the equivalent of 98 gates on a monolithic chip measuring only 90 by 110 mils. The register file is organized as 4 words of 4 bits each and separate on-chip decoding is provided for addressing the four word locations to either write-in or retrieve data. This permits simultaneous writing into one location and reading from another word location.

Four data inputs are available which are used to supply the 4-bit word to be stored. Location of the word is determined by the write address inputs A and B in conjunction with a write-enable signal. Data applied at the inputs should be in its true form. That is, if a high-level signal is desired from the output, a high-level is applied at the data input for that particular bit location. The latch inputs are arranged so that new data will be accepted only if both internal address gate inputs are high. When this condition exists, data at the D input is transferred to the latch output. When the write-enable input, G_W, is high, the data inputs are inhibited and their levels can cause no change in the information stored in the internal latches. When the read-enable input, G_R, is high, the data outputs are inhibited and remain high.

The individual address lines permit direct acquisition of data stored in any four of the latches. Four individual decoding gates are used to complete the address for reading a word. When the read address is made in conjunction with the read-enable signal, the word appears at the four outputs.

This arrangement (data-entry addressing separate from data-read addressing and individual sense line) eliminates recovery times, permits simultaneous reading and writing, and is limited in speed only by the write time (45 nanoseconds maximum) and the read time (35 nanoseconds maximum). The register file has a nondestructive readout in that data is not lost when addressed.

All inputs are buffered to lower the drive requirements to one normalized load, and input-clamping diodes minimize switching transients to simplify system design. High-speed, double-ended AND-OR-INVERT gates are employed for the read-address function and drive high-sink-current, open-collector outputs. Up to 246 of these outputs may be wire-AND connected for increasing the capacity up to 1024 words. Any number of these registers may be paralleled to provide n-bit word length.

2/ Device will fanout in both high- and low-levels to the specified number of inputs of the same device as that being tested.

3/ With respect to write-enable.

4/ Write select setup time will protect the data written into the previous address. If the protection of data in the previous address is not required, write select setup time can be ignored as any address selection sustained for the final 30 ns of the write-enable pulse and during the write select hold time will result in data being written with that location. Depending on the duration of the input conditions, one or a number of previous addresses may have been written into.

5/ Latch time is the time required for the internal output of the latch to assume the state of the new data. This is important only when attempting to read from a location immediately after that location has received new data.

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2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagram. The logic diagram shall be as specified on figure 2.

3.2.3 Truth tables and functional description. The truth tables or functional description shall be as specified on figure 3.

3.2.4 Case outlines. Case outlines shall be as specified in MIL-M-38510 and in 1.2.3 herein.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container. The "JAN" or "J" certification mark shall not be used.

3.7 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have a manufacturer certification in accordance with MIL-M-38510 for at least one line. Not necessarily the line producing the device type described herein.

3.8 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/	Limits Min	Limits Max	Unit
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 16 mA		0.4	V
Maximum collector cut-off current	I_{CEX}	V_{CC} = 4.5 V, V_O = 5.5 V		30	µA
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = 12 mA		-1.5	V
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	-0.7	-1.6	mA
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V		40	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		100	µA
Supply current	I_{CC}	V_{CC} = 5.5 V		140	mA
Propagation delay times	t_{PLH1}	Read-enable to any output, V_{CC} = 5.0 V, R_L = 400Ω, C_L = 50 pF	5	18	ns
	t_{PHL1}		5	36	ns
	t_{PLH2}	Write-enable to any output, V_{CC} = 5.0 V, R_L = 400Ω, C_L = 50 pF	5	48	ns
	t_{PHL2}		5	54	ns
	t_{PLH3}	Read address to any output, V_{CC} = 5.0 V, R_L = 400Ω, C_L = 50 pF	5	42	ns
	t_{PHL3}		5	54	ns
	t_{PLH4}	Data to any output, V_{CC} = 5.0 V, R_L = 400Ω, C_L = 50 pF	5	36	ns
	t_{PHL4}		5	54	ns

1/ Complete terminal conditions are specified in table III.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) Class B devices
Interim electrical test parameters (pre burn-in) (method 5004)	1
Final electrical test parameters (method 5004)	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,9
Group C end-point electrical parameters (method 5005)	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3
Additional electrical subgroups for group C periodic inspection	10,11

*PDA applies to subgroup 1 (see 4.2c).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510, and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.6) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

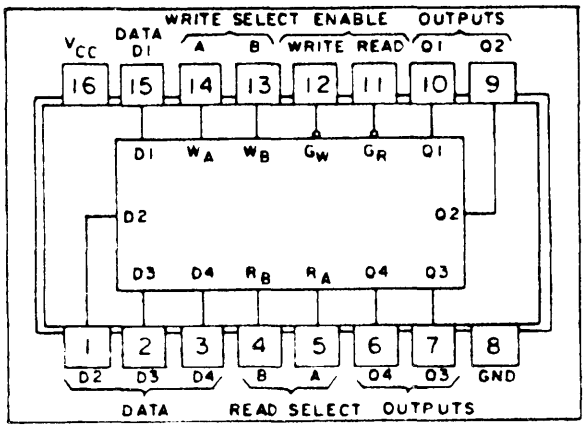
4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical paramters shall be as specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical test parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

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CASES E AND F

FIGURE 1. Terminal connections.

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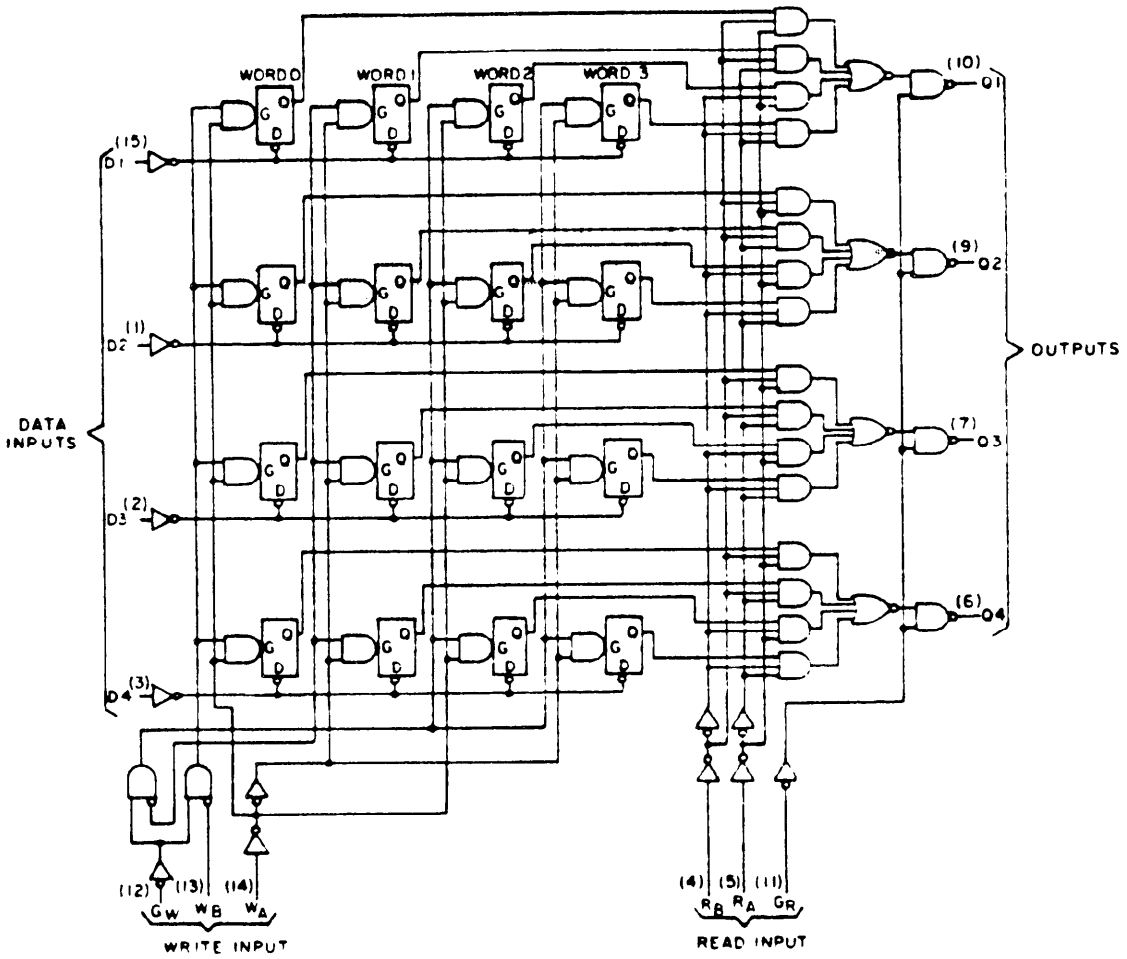


FIGURE 2. Logic diagram.

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WRITE FUNCTION TABLE (SEE NOTES A, B, AND C)

WRITE INPUTS			WORD			
W_B	W_A	G_W	0	1	2	3
L	L	L	Q = D	Q_0	Q_0	Q_0
L	H	L	Q_0	Q = D	Q_0	Q_0
H	L	L	Q_0	Q_0	Q = D	Q_0
H	H	L	Q_0	Q_0	Q_0	Q = D
X	X	H	Q_0	Q_0	Q_0	Q_0

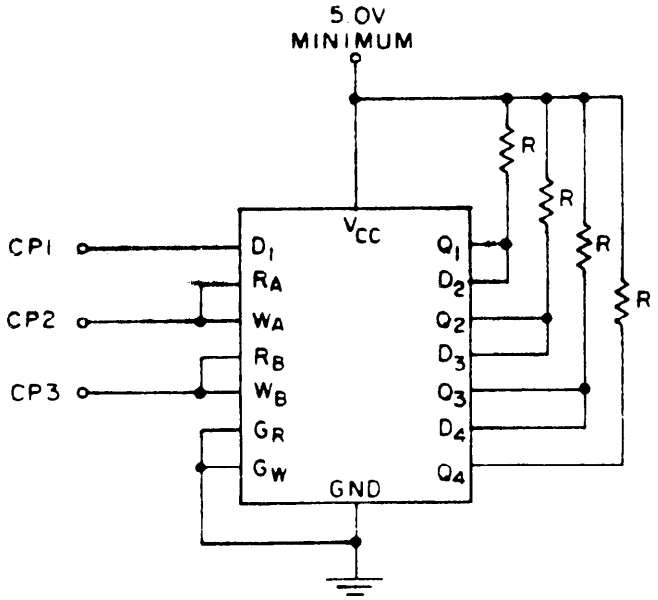
READ FUNCTION TABLE (SEE NOTES A AND D)

READ INPUTS			OUTPUTS			
R_B	R_A	G_R	Q_1	Q_2	Q_3	Q_4
L	L	L	W0B1	W0B2	W0B3	W0B4
L	H	L	W1B1	W1B2	W1B3	W1B4
H	L	L	W2B1	W2B2	W2B3	W2B4
H	H	L	W3B1	W3B2	W3B3	W3B4
X	X	H	H	H	H	H

NOTES:
A. H = high level, L = low level, X = irrelevant.
B. (Q = D) = The four selected internal flip-flop outputs will assume the states applied to the four external data inputs.
C. Q_0 = The level of Q before the indicated input conditions were established.
D. W0B1 = The first bit of word 0, etc.

FIGURE 3. Truth tables.

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NOTES:
1. R = 270Ω ±5%.
2. All clock pulse generators have the following characteristics: V_{gen} = 3.0 V minimum, 50% duty cycle, t_{THL} and $t_{TLH} \leq 10$ ns and frequency as follows.

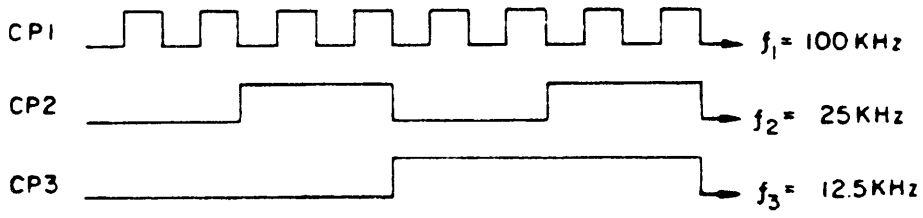
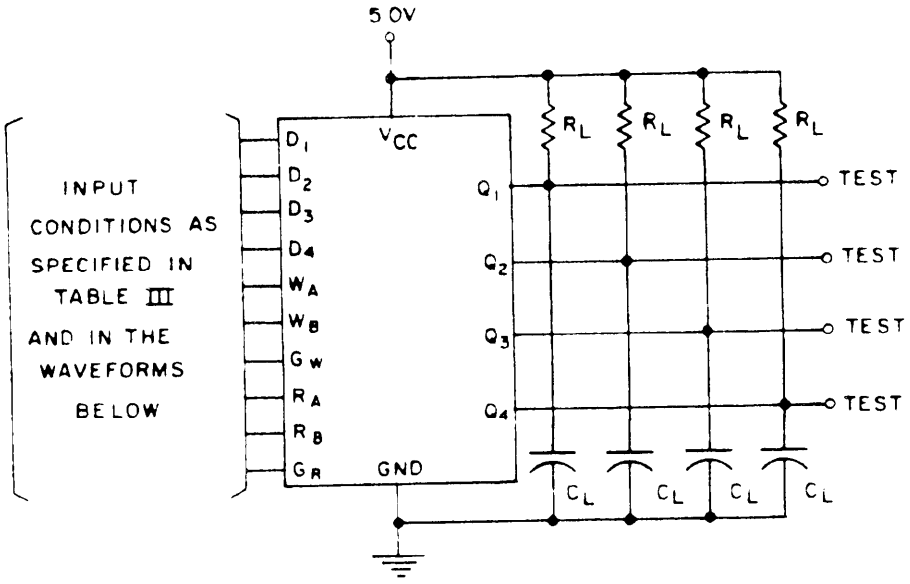
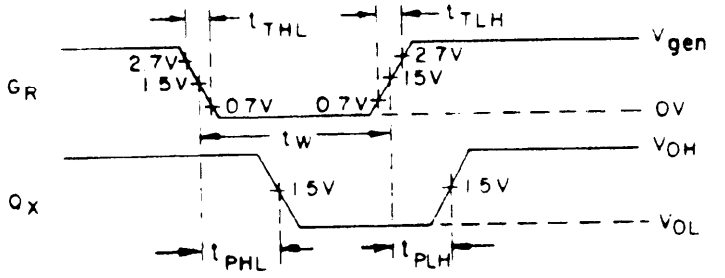


FIGURE 4. Burn-in and life test circuits.

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A. READ-ENABLE TO ANY OUTPUT (t_{PHL1} & t_{PLH1})

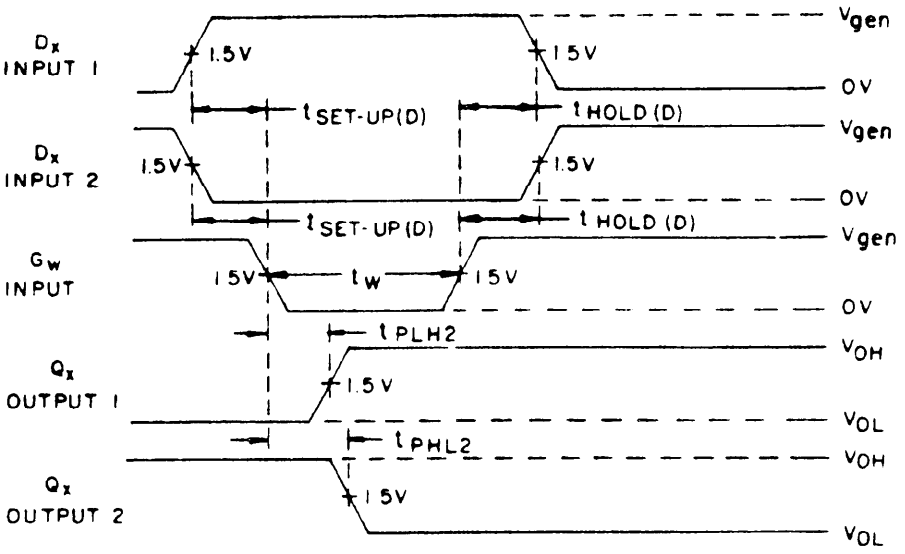


NOTES for A:
1. RL = 400Ω ±5%.
2. CL = 50 pF ±10%, including probe and jig capacitance.
3. Read enable (GR) pulse characteristics shall be as follows: Vgen = 3.0 V minimum, t_{THL} and $t_{TLH} \leq$ 10 ns, tW = 25 ns, and PRR = 1 MHz.

FIGURE 5. Switching circuit and waveforms.

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B. WRITE ENABLE TO ANY OUTPUT (t_{PLH2} AND t_{PHL2})

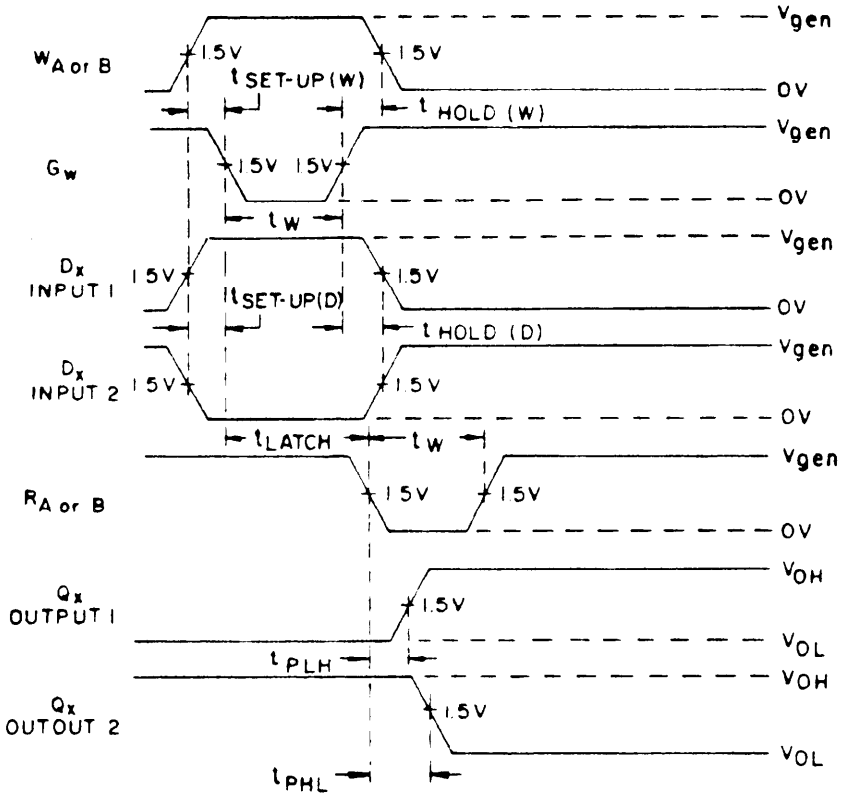


NOTES for B:
1. All pulse generators have the following characteristics: t_{TLH} (0.7 V to 2.7 V) and t_{THL} (2.7 V to 0.7 V) $\leq$ 10 ns and $V_{gen} \geq$ 3.0 V.
2. PRR for "D" pulse generator shall be 500 kHz. PRR for "GW" pulse generator shall be 1 MHz.
3. $t_{SETUP(D)}$ = 10 ns, $t_{HOLD(D)}$ = 15 ns and t_W = 25 ns.

FIGURE 5. Switching circuit and waveforms - Continued.

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C READ-ADDRESS (R_A or R_B) TO ANY OUTPUT (t_{PHL3} AND t_{PLH3})

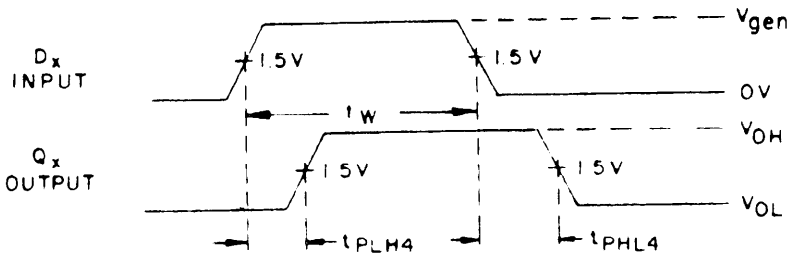


NOTES for C:
1. All pulse generators have the following characteristics: t_{TLH} (0.7 V to 2.7 V) and t_{THL} (2.7 V to 0.7 V) $\leq$ 10 ns, and $V_{gen} \geq 3.0$ V.
2. $t_{SETUP(W)}$ = 15 ns, $t_{HOLD(W)}$ = 5 ns, $t_{SETUP(D)}$ = 10 ns, $t_{HOLD(D)}$ = 15 ns and t_W = 25 ns.
3. PRR = 1 MHz for W_A or B and G_W; PRR = 500 kHz for DX and R_A or B.

FIGURE 5. Switching circuit and waveforms - Continued.

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D. DATA INPUT TO Q OUTPUT (t_{PLH4} AND t_{PHL4})



NOTES for D:
1. Data pulse generator has the following characteristics: t_{TLH} (0.7 V to 2.7 V) and t_{THL} (2.7 V to 0.7 V) $\leq$ 10 ns, Vgen $\geq$ 3.0 V, tW $\leq$ 100 ns and PRR = 1 MHz.

FIGURE 5. Switching circuit and waveforms - Continued.

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TABLE III Group A inspection for device type 01.
Terminal conditions (pins not designated are high level, low level or open)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 D_2	2 D_3	3 D_4	4 R_B	5 R_A	6 Q_4	7 Q_3	8 GND	9 Q_2	10 Q_1	11 G_R	12 G_W	13 W_B	14 W_A	15 D_1	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	V_{OL}	3007	1	0.8 V			0.8 V	0.8 V			GND		16 mA	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	4.5 V	Q_1		0.4	V
			2				0.8 V	2.0 V				16 mA				0.8 V	2.0 V			Q_2			
			3		0.8 V		2.0 V	0.8 V		16 mA						2.0 V	0.8 V			Q_3			
			4			0.8 V	2.0 V	2.0 V	16 mA							2.0 V	2.0 V			Q_4			
	I_{CEX}	3008	5				0.8 V	0.8 V					5.5 V			0.8 V	0.8 V	2.0 V		Q_1		30	μA
			6	2.0 V			0.8 V	2.0 V				5.5 V				0.8 V	2.0 V			Q_2			
			7		2.0 V		2.0 V	0.8 V		5.5 V						2.0 V	0.8 V			Q_3			
			8			2.0 V	2.0 V	2.0 V	5.5 V							2.0 V	2.0 V			Q_4			
	V_{IC}		9															-12 mA		D_1		-1.5	V
			10	-12 mA																D_2			
			11		-12 mA															D_3			
			12			-12 mA														D_4			
			13					-12 mA												R_A			
			14				-12 mA													R_B			
			15											-12 mA						G_R			
			16												-12 mA					G_W			
			17														-12 mA			W_A			
			18													-12 mA				W_B			
	I_{IL}	3009	19															0.4 V	5.5 V	D_1	-0.7	-1.6	mA
			20	0.4 V																D_2			
			21		0.4 V															D_3			
			22			0.4 V														D_4			
			23					0.4 V												R_A			
			24				0.4 V													R_B			
			25											0.4 V						G_R			
			26												0.4 V					G_W			
			27														0.4 V			W_A			
			28													0.4 V				W_B			
	I_{IH1}	3010	29															2.4 V		D_1	0	40	μA
			30	2.4 V																D_2			
			31		2.4 V															D_3			
			32			2.4 V														D_4			
			33					2.4 V												R_A			
			34				2.4 V													R_B			
			35											2.4 V						G_R			
			36												2.4 V					G_W			
			37														2.4 V			W_A			
			38													2.4 V				W_B			
	I_{IH2}		39															5.5 V		D_1		100	
			40	5.5 V																D_2			
			41		5.5 V															D_3			
			42			5.5 V														D_4			
			43					5.5 V												R_A			
			44				5.5 V													R_B			
			45											5.5 V						G_R			
			46												5.5 V					G_W			
			47														5.5 V			W_A			
			48													5.5 V				W_B			
	I_{CC}	3005	49	4.5 V	4.5 V	4.5 V	GND	GND						4.5 V	4.5 V	GND	GND	4.5 V		V_{CC}		140	mA
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

MIL-M-38510/10A

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated are high level, low level or open)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 D_2	2 D_3	3 D_4	4 R_B	5 R_A	6 Q_4	7 Q_3	8 GND	9 Q_2	10 Q_1	11 G_R	12 G_W	13 W_B	14 W_A	15 D_1	16 V_{CC}	Meas. terminal	Test Limits Min	Max	Unit
7 T_C = 25°C	(Truth table test)		50	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	L	L	GND	L	L	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	5.0 V	See note at the end of this table			
			51	2.0 V	2.0 V	2.0 V		0.8 V	H	H		H	H	0.8 V	0.8 V		0.8 V	2.0 V					
			52	2.0 V	2.0 V	2.0 V		0.8 V	H	H		H	H	2.0 V	2.0 V		0.8 V	2.0 V					
			53	0.8 V	0.8 V	0.8 V		2.0 V	H	H		H	H	2.0 V	2.0 V		2.0 V	0.8 V					
			54	0.8 V	0.8 V	0.8 V		2.0 V	L	L		L	L	0.8 V	0.8 V		2.0 V	0.8 V					
			55	2.0 V	2.0 V	2.0 V		2.0 V	H	H		H	H	0.8 V	0.8 V		2.0 V	2.0 V					
			56	2.0 V	2.0 V	2.0 V		2.0 V	H	H		H	H	2.0 V	2.0 V		2.0 V	2.0 V					
			57	2.0 V	0.8 V	2.0 V	2.0 V	0.8 V	H	H		H	H	2.0 V	2.0 V	2.0 V	0.8 V	0.8 V					
			58	2.0 V	0.8 V	2.0 V		0.8 V	H	L		H	L	0.8 V	0.8 V		0.8 V	0.8 V					
			59	0.8 V	2.0 V	0.8 V		0.8 V	L	H		L	H	0.8 V	0.8 V		0.8 V	2.0 V					
			60	0.8 V	2.0 V	0.8 V		0.8 V	H	H		H	H	2.0 V	2.0 V		0.8 V	2.0 V					
			61	0.8 V	0.8 V	0.8 V		2.0 V	H	H		H	H	2.0 V	2.0 V		2.0 V	0.8 V					
			62	0.8 V	0.8 V	0.8 V		2.0 V	L	L		L	L	0.8 V	0.8 V		2.0 V	0.8 V					
			63	2.0 V	2.0 V	2.0 V		2.0 V	H	H		H	H	0.8 V	0.8 V		2.0 V	2.0 V					
			64	2.0 V	2.0 V	2.0 V		2.0 V	H	H		H	H	2.0 V	2.0 V		2.0 V	2.0 V					
			65	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	H	H		H	H	2.0 V	2.0 V	0.8 V	0.8 V	0.8 V					
			66	0.8 V	0.8 V	0.8 V		0.8 V	L	L		L	L	0.8 V	0.8 V		0.8 V	0.8 V					
			67	0.8 V	0.8 V	0.8 V		0.8 V	H	H		H	H	2.0 V	2.0 V		0.8 V	0.8 V					
			68	2.0 V	2.0 V	2.0 V		0.8 V	H	H		H	H	2.0 V	2.0 V		2.0 V	2.0 V					
			69	2.0 V	2.0 V	2.0 V		0.8 V	L	L		L	L	0.8 V	2.0 V		2.0 V	2.0 V					
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																						
9 T_C = 25°C	t_{PLH1}	3003 (Fig. 5)	70	GND	GND	GND	GND	GND		OUT	GND	OUT	OUT	IN	GND	GND	GND	GND	5.0 V	G_R to Q_1	5	15	ns
			71																	G_R to Q_2			
			72							OUT										G_R to Q_3			
			73						OUT											G_R to Q_4			
	t_{PHL1}		74										OUT							G_R to Q_1		30	
			75									OUT								G_R to Q_2			
			76							OUT										G_R to Q_3			
			77						OUT											G_R to Q_4			
	t_{PLH2}		78										OUT	GND	IN	GND	GND	IN1		G_W to Q_1		40	
			79	IN1								OUT								G_W to Q_2			
			80		IN1					OUT										G_W to Q_3			
			81			IN1			OUT											G_W to Q_4			
			82					5.0 V					OUT				5.0 V	IN1		G_W to Q_1			
			83	IN1								OUT								G_W to Q_2			
			84		IN1					OUT										G_W to Q_3			
			85			IN1			OUT											G_W to Q_4			
			86				5.0 V	GND					OUT			5.0 V	GND	IN1		G_W to Q_1			
			87	IN1								OUT								G_W to Q_2			
			88		IN1					OUT										G_W to Q_3			
			89			IN1			OUT											G_W to Q_4			
			90					5.0 V					OUT					IN1		G_W to Q_1			
			91	IN1								OUT								G_W to Q_2			
			92		IN1					OUT										G_W to Q_3			
			93			IN1			OUT											G_W to Q_4			

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated are high level, low level or open)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 D_2	2 D_3	3 D_4	4 R_B	5 R_A	6 Q_4	7 Q_3	8 GND	9 Q_2	10 Q_1	11 G_R	12 G_W	13 W_B	14 W_A	15 D_1	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
9 T_C = 25°C	t_{PHL2}	3003 (Fig 5)	94	IN2	IN2	IN2	GND	GND	OUT	OUT	GND	OUT	OUT	GND	IN	GND	GND	IN2	5.0 V	G_W to Q_1	5	45	ns
			95																	G_W to Q_2			
			96																	G_W to Q_3			
			97																	G_W to Q_4			
			98					5.0 V					OUT				5.0 V	IN2		G_W to Q_1			
			99	IN2								OUT								G_W to Q_2			
			100		IN2					OUT										G_W to Q_3			
			101			IN2			OUT											G_W to Q_4			
			102				5.0 V	GND					OUT			5.0 V	GND	IN2		G_W to Q_1			
			103	IN2								OUT								G_W to Q_2			
			104		IN2					OUT										G_W to Q_3			
			105			IN2			OUT											G_W to Q_4			
			106					5.0 V									5.0 V	IN2		G_W to Q_1			
			107	IN2																G_W to Q_2			
			108		IN2															G_W to Q_3			
			109			IN2														G_W to Q_4			
	t_{PLH3}		110	IN1	IN1	IN1	GND	IN	OUT1	OUT1		OUT1	OUT1			GND	IN	IN1		R_A to Q_1		35	
			111																	R_A to Q_2			
			112																	R_A to Q_3			
			113																	R_A to Q_4			
	t_{PHL3}		114	IN2	IN2	IN2	IN	GND	OUT2	OUT2		OUT2	OUT2			IN	GND	IN2		R_B to Q_1		45	
			115																	R_B to Q_2			
			116																	R_B to Q_3			
			117																	R_B to Q_4			
	t_{PLH4}		118	IN	IN	IN	GND		OUT	OUT		OUT	OUT		GND	GND		IN		D_1 to Q_1		30	
			119																	D_2 to Q_2			
			120																	D_3 to Q_3			
			121																	D_4 to Q_4			
	t_{PHL4}		122	IN	IN	IN			OUT	OUT		OUT	OUT					IN		D_1 to Q_1		45	
			123																	D_2 to Q_2			
			124																	D_3 to Q_3			
			125																	D_4 to Q_4			
10 T_C = 125°C	t_{PLH1}		126	GND	GND	GND			OUT	OUT		OUT	OUT	IN				GND		G_R to Q_1		18	
			127																	G_R to Q_2			
			128																	G_R to Q_3			
			129																	G_R to Q_4			
	t_{PHL1}		130						OUT	OUT		OUT	OUT							G_R to Q_1		36	
			131																	G_R to Q_2			
			132																	G_R to Q_3			
			133																	G_R to Q_4			
	t_{PLH2}		134	IN1	IN1	IN1			OUT	OUT		OUT	OUT	GND	IN			IN1		G_W to Q_1		48	
			135																	G_W to Q_2			
			136																	G_W to Q_3			
			137																	G_W to Q_4			

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated are high level, low level or open)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 D_2	2 D_3	3 D_4	4 R_B	5 R_A	6 Q_4	7 Q_3	8 GND	9 Q_2	10 Q_1	11 G_R	12 G_W	13 W_B	14 W_A	15 D_1	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
10 T_C 125°C	t_{PLH2}	3003 (Fig 5)	138				GND	5.0 V			GND		OUT	GND	IN	GND	5.0 V	IN1	5.0 V	G_W to Q_1	5	48	ns
			139	IN1								OUT								G_W to Q_2			
			140		IN1					OUT										G_W to Q_3			
			141			IN1			OUT											G_W to Q_4			
			142				5.0 V	GND					OUT			5.0 V	GND	IN1		G_W to Q_1			
			143	IN1								OUT								G_W to Q_2			
			144		IN1					OUT										G_W to Q_3			
			145			IN1			OUT											G_W to Q_4			
			146					5.0 V					OUT				GND	IN1		G_W to Q_1			
			147	IN1								OUT								G_W to Q_2			
			148		IN1					OUT										G_W to Q_3			
			149			IN1			OUT											G_W to Q_4			
	t_{PHL2}		150				GND	GND					OUT			GND	GND	IN2		G_W to Q_1		54	
			151	IN2								OUT								G_W to Q_2			
			152		IN2					OUT										G_W to Q_3			
			153			IN2			OUT											G_W to Q_4			
			154					5.0 V					OUT				5.0 V	IN2		G_W to Q_1			
			155	IN2								OUT								G_W to Q_2			
			156		IN2					OUT										G_W to Q_3			
			157			IN2			OUT											G_W to Q_4			
			158				5.0 V	GND					OUT			5.0 V	GND	IN2		G_W to Q_1			
			159	IN2								OUT								G_W to Q_2			
			160		IN2					OUT										G_W to Q_3			
			161			IN2			OUT											G_W to Q_4			
			162					5.0 V					OUT				5.0 V	IN2		G_W to Q_1			
			163	IN2								OUT								G_W to Q_2			
			164		IN2					OUT										G_W to Q_3			
			165			IN2			OUT											G_W to Q_4			
	t_{PLH3}		166				GND	IN					OUT1			GND	IN	IN1		R_A to Q_1		42	
			167	IN1								OUT1								R_A to Q_2			
			168		IN1					OUT1										R_A to Q_3			
			169			IN1			OUT1											R_A to Q_4			
	t_{PHL3}		170				IN	GND					OUT2			IN	GND	IN2		R_B to Q_1		54	
			171	IN2								OUT2								R_B to Q_2			
			172		IN2					OUT2										R_B to Q_3			
			173			IN2			OUT2											R_B to Q_4			
	t_{PLH4}		174				GND						OUT		GND	GND		IN		D_1 to Q_1		36	
			175	IN								OUT								D_2 to Q_2			
			176		IN					OUT										D_3 to Q_3			
			177			IN			OUT											D_4 to Q_4			
	t_{PHL4}		178										OUT					IN		D_1 to Q_1		54	
			179	IN								OUT								D_2 to Q_2			
			180		IN					OUT										D_3 to Q_3			
			181			IN			OUT											D_4 to Q_4			
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																						

NOTE: Output voltages shall be either:
(a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
(b) H ≥ 1.0 volt and L ≤ 1.0 volt when using a high speed checker single comparator.

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c. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging of microcircuits shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity, if applicable.

e. Requirements for packaging and packing.

f. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal)
I_{IN}	Current flowing into an input terminal
T_C	Case temperature
V_{IN}	Voltage level at an input terminal

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.6 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.4) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54170

6.8 Ordering guidance. Since the qualification and certification requirements have been removed from the specification, orders may be placed immediately.

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Preparing activity:
Air Force - 17

(Project 5962-0591-1)

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/18A	MICROCIRCUITS, DIGITAL, TTL, REGISTER FILE, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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